

GEOGRAPHY

PHYSICAL GEOGRAPHY

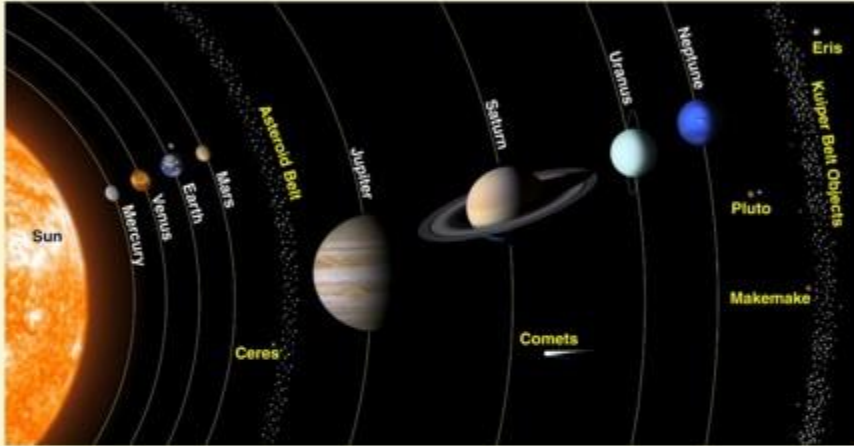


MODULE -7

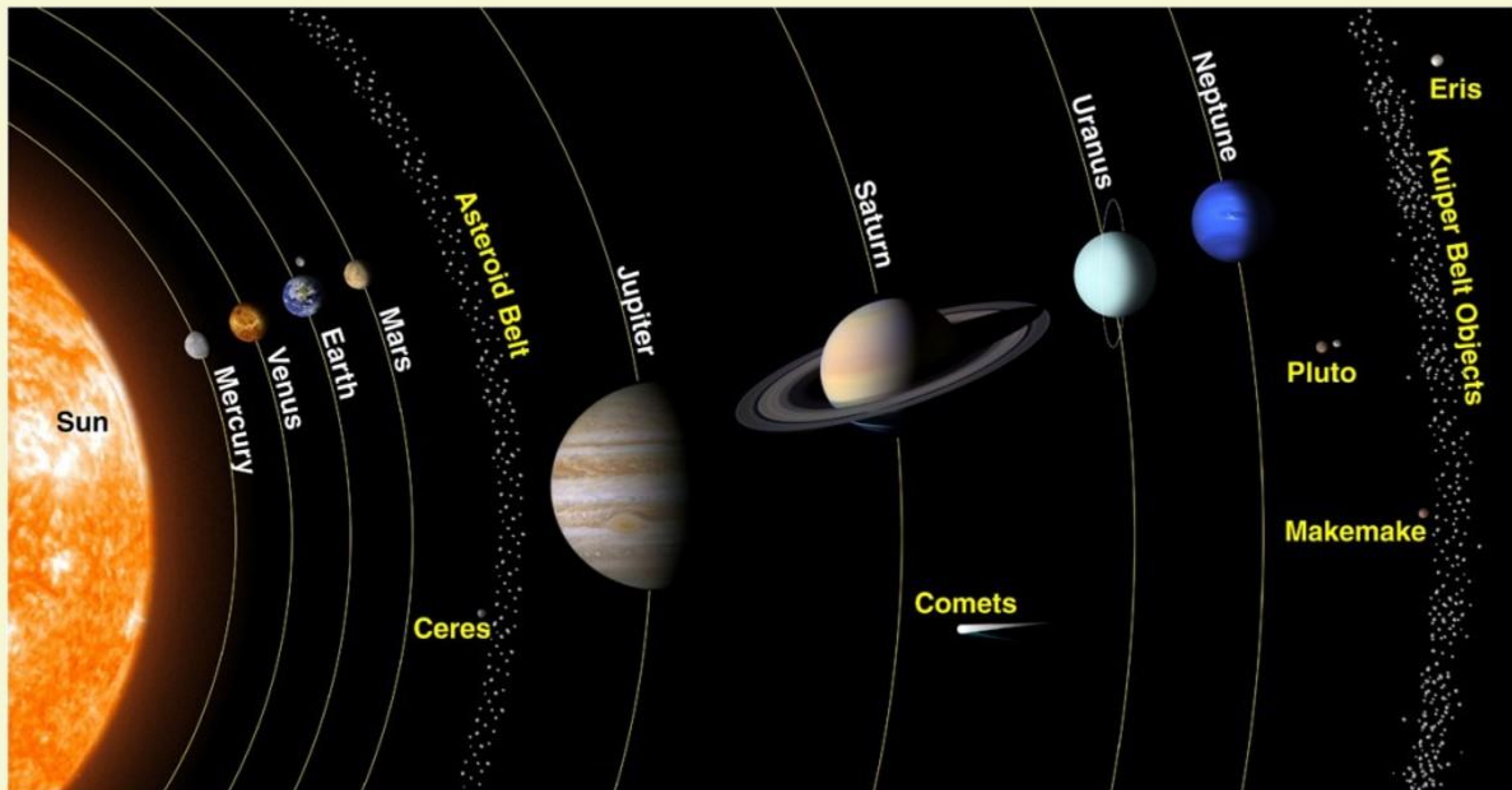
- SOLAR SYSTEM AND EARTH
- SPHERICITY OF EARTH
- WHY THE EARTH ROTATES?
- WHY THE EARTH REVOLVES?



SOLAR SYSTEM AND EARTH

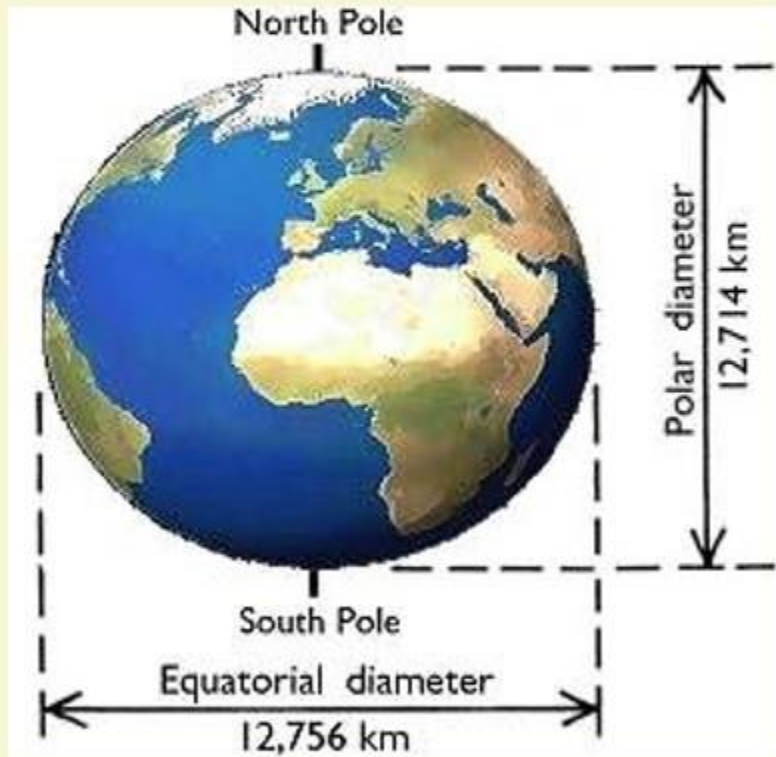
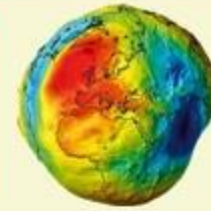


- 1** The solar system comprises the Sun and the 8 planets
- 2** All the planets revolve around the Sun in elliptical orbits.
- 3** Just like Earth, they shine only by the reflected light of the Sun.
- 4** The Sun is about 3,00,000 times as massive as the Earth.
- 5** All over the surface of the Sun, there are violent gases, which leap up as glowing flames like a volcano in eruption.





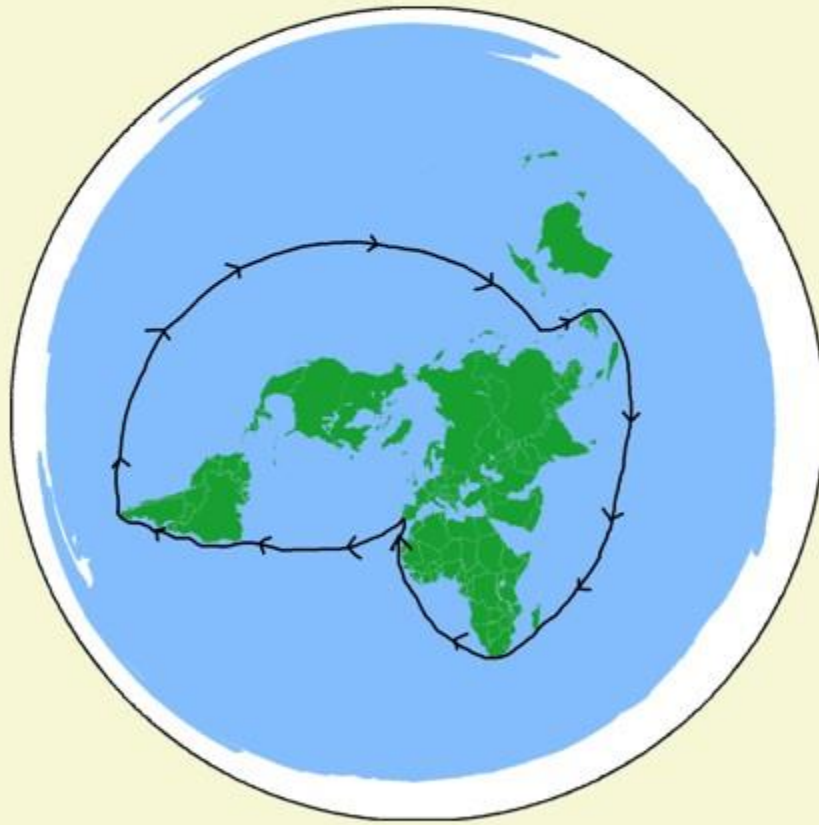
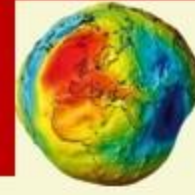
SHAPE OF THE EARTH



- 1 Earth has got equatorial circumference of **40,070** km and its polar circumference is less by **133.5** km.
- 2 Its equatorial diameter is **12,756** km and its polar diameter is shorter by **41.84** km.
- 3 Earth is not a perfect sphere, it is little flattened at both ends.
- 4 Hence, it is called a **Geoid**.
- 5 The spherical shape of the Earth is also masked by the intervening high lands and oceans on its surface.



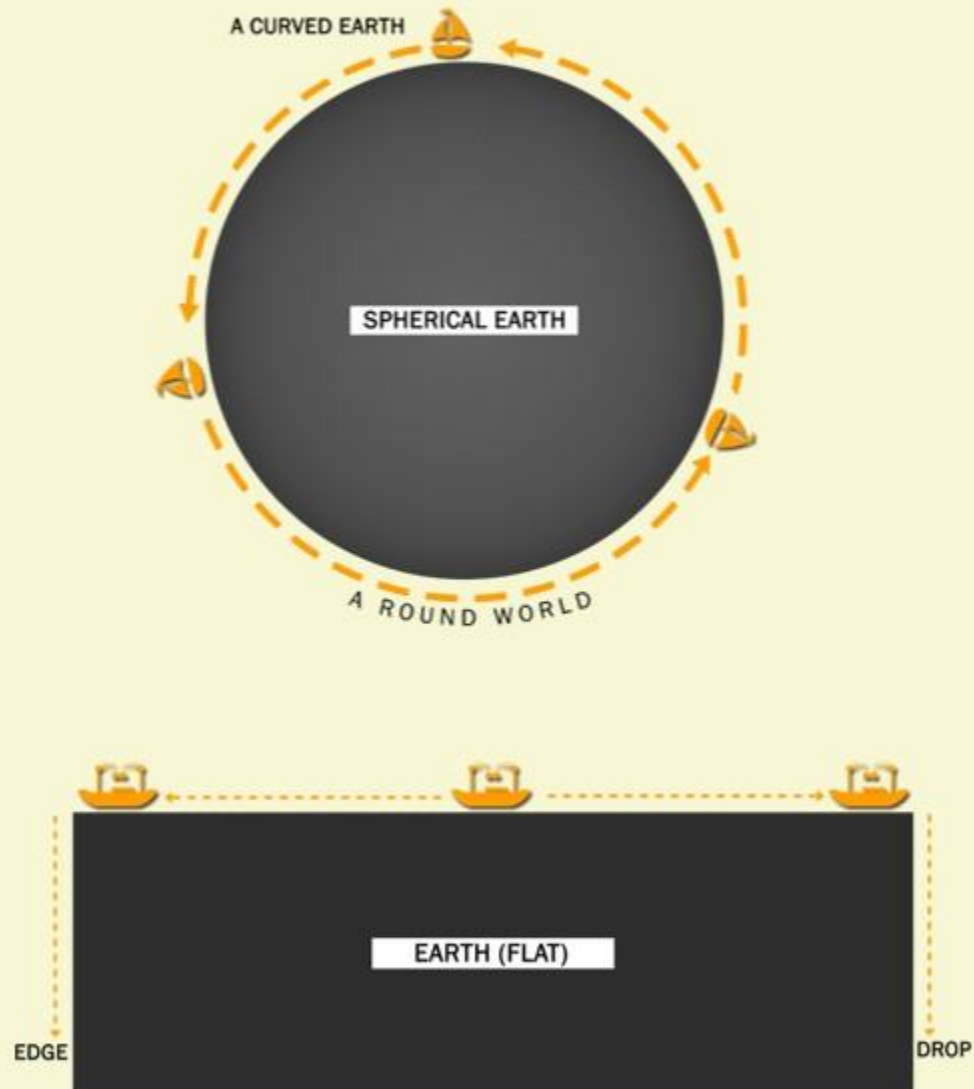
EVIDENCE TO SHOW EARTH'S SPHERICITY



1

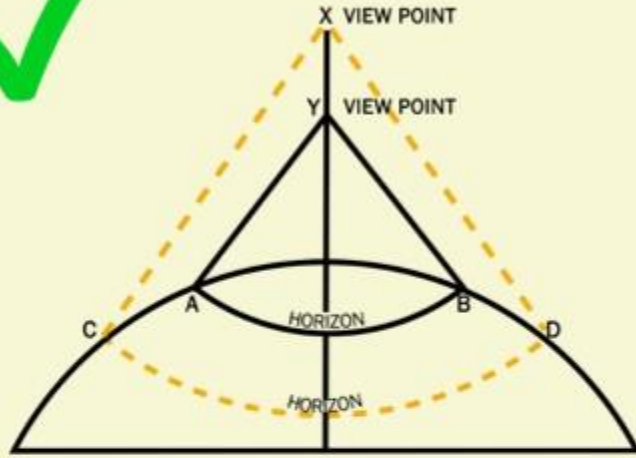
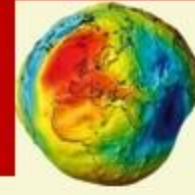
CIRCUM-NAVIGATION OF THE EARTH

- The first voyage around the world by Ferdinand Magellan from 1519 to 1522 proved beyond doubt that the Earth is spherical.
- Modern air routes and ocean navigation are all based on the assumption that the Earth is round.



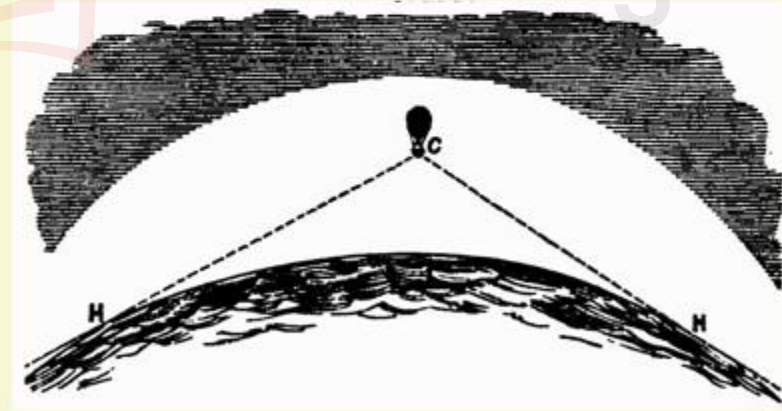
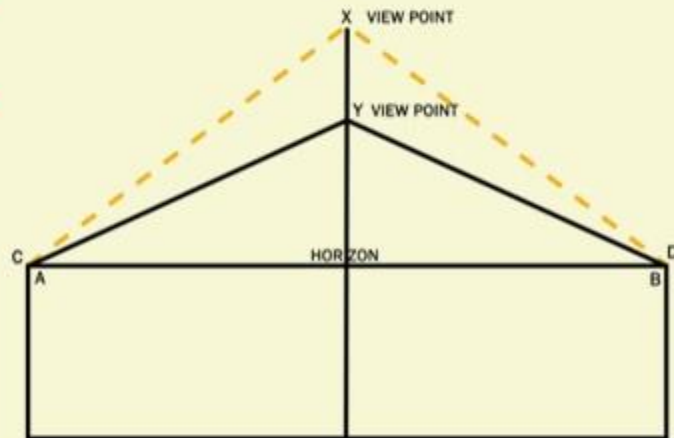


EVIDENCE TO SHOW EARTH'S SPHERICITY



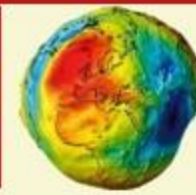
2 THE CIRCULAR HORIZON

- With the increasing altitude, the circular horizon is widening and this could only be possible on a spherical body.

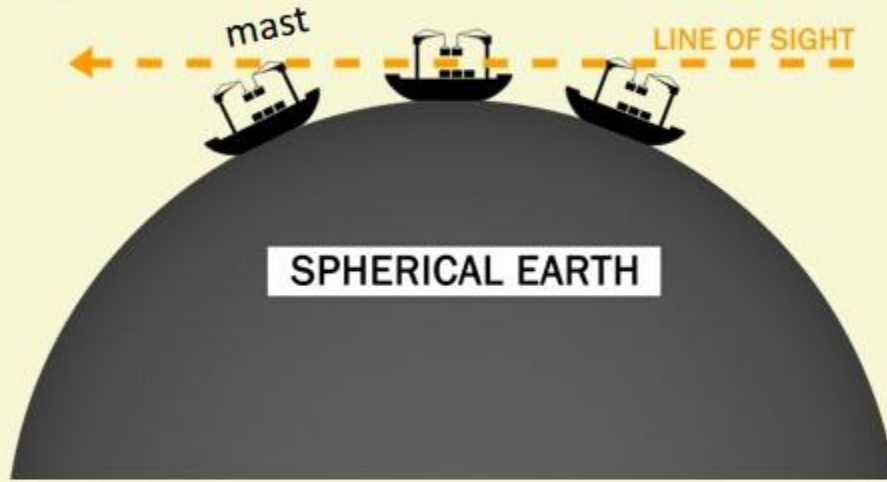




EVIDENCE TO SHOW EARTH'S SPHERICITY



A CURVED EARTH



LINE OF SIGHT



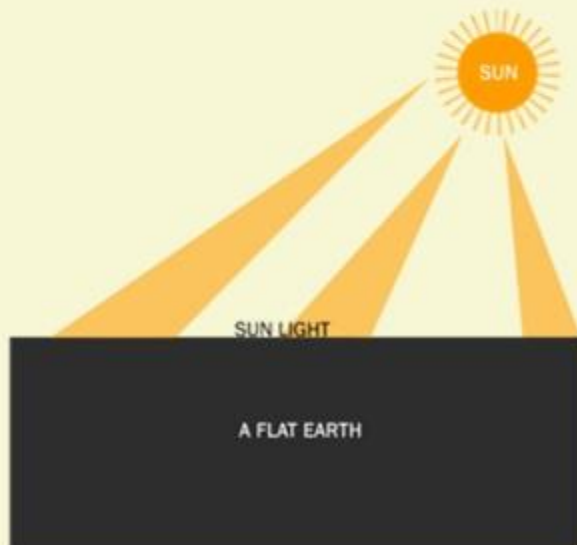
3

VISIBILITY OF THE SHIP

- When a ship appears over the distant horizon, the top of the mast is seen first before the hull.
- When it leaves harbour, its disappearance over the curved surface is equally gradual.
- It is possible only when the Earth is spherical.



EVIDENCE TO SHOW EARTH'S SPHERICITY



4

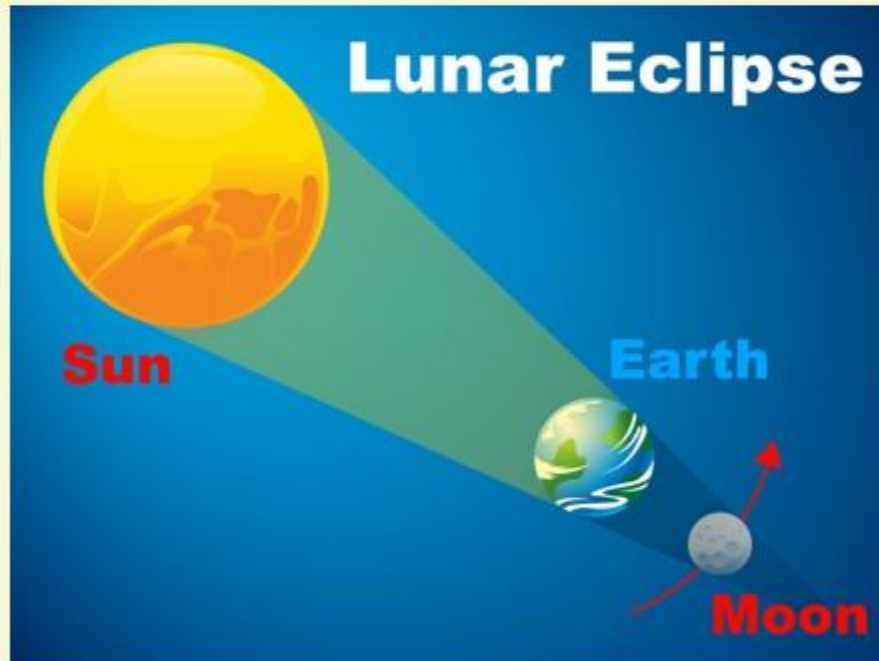
SUNRISE AND SUNSET



- The Sun rises and sets at different times in different places.
- As the Earth rotates from West to East, places in the East see the Sun earlier than those in the West.
- Had the Earth was flat, the whole world would have sunrise and sunset at the same time.



EVIDENCE TO SHOW EARTH'S SPHERICITY



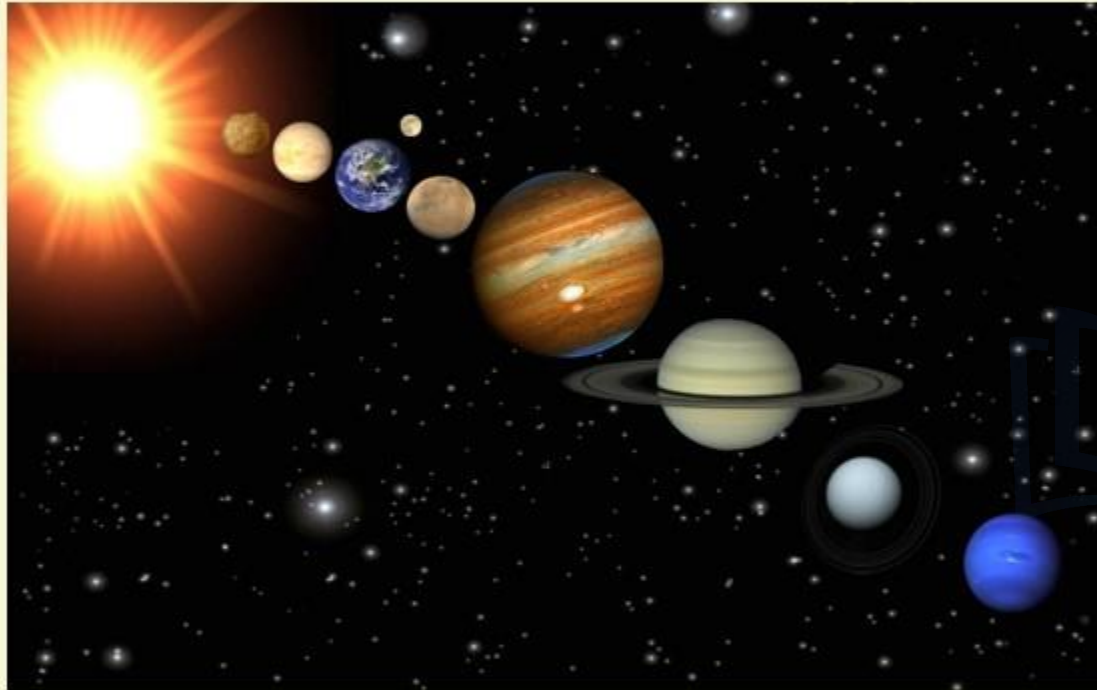
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THE LUNAR ECLIPSE

- The shadow caused by the Earth on the Moon during the Lunar eclipse is always circular.
- Only a Sphere can cast such a circular shadow.



EVIDENCE TO SHOW EARTH'S SPHERICITY



6

PLANETARY BODIES ARE SPHERICAL

- The observations from telescope reveal that the planetary bodies, the Sun, Moon, satellites and stars have circular outlines and they are strictly spheres and Earth cannot be an exception.



EVIDENCE TO SHOW EARTH'S SPHERICITY



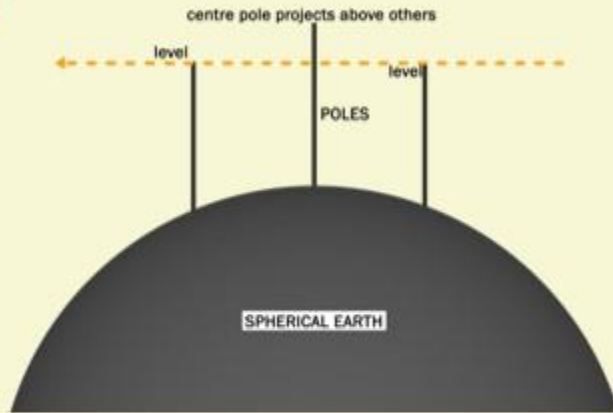
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AERIAL PHOTOGRAPHS

- Pictures taken from high altitudes by rockets and satellites show clearly the curved edge of the Earth and this is the most convincing and up-to-date proof of the Earth's sphericity.



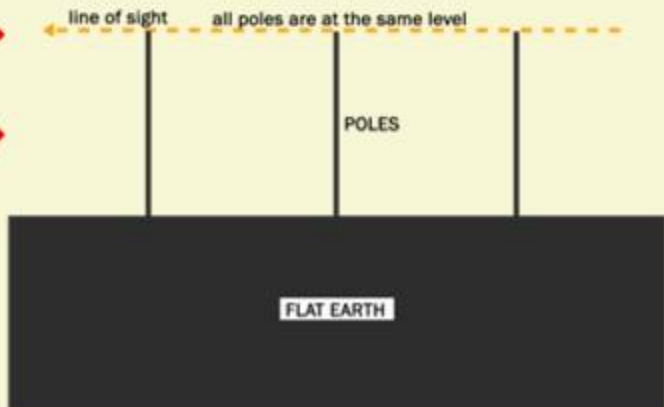
EVIDENCE TO SHOW EARTH'S SPHERICITY



8

DRIVING POLES ON LEVEL GROUND ON A CURVED PATH

- Engineers when driving poles of equal length at regular intervals on the ground have found that they do not have a perfect horizontal level.
- The central pole normally projects slightly above the poles at the either end, because of the curvature of the Earth.









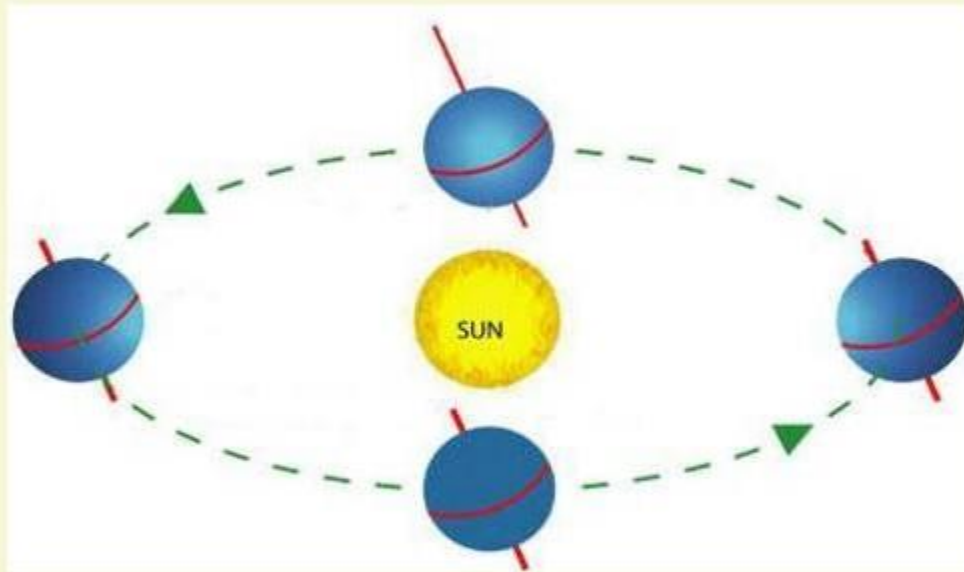
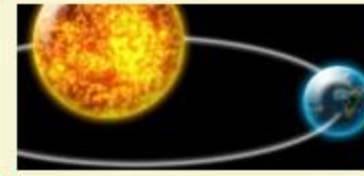
ROTATION OF THE EARTH



- 1 Earth rotates on its own axis.
- 2 Hence, one portion of the Earth's surface comes towards the rays of the Sun and experiences day light.
- 3 The other portion is away from the Sun's rays and that will be darkness.
- 4 The Earth rotates from west to east and hence, every part of the Earth's surface will be brought under the Sun at sometime or the other.



REVOLUTION OF THE EARTH



- 1** When Earth revolves around the Sun, it revolves on an elliptical orbit at a speed of 18.5 miles/s.
- 2** One complete revolution takes around $365\frac{1}{4}$ days or nearly a year.
- 3** Normal year is taken as 365 days and leap year is taken as 366 days to adjust for around $\frac{1}{4}^{\text{th}}$ extra day every year.

DIFFERENCE BETWEEN



ROTATION OF EARTH

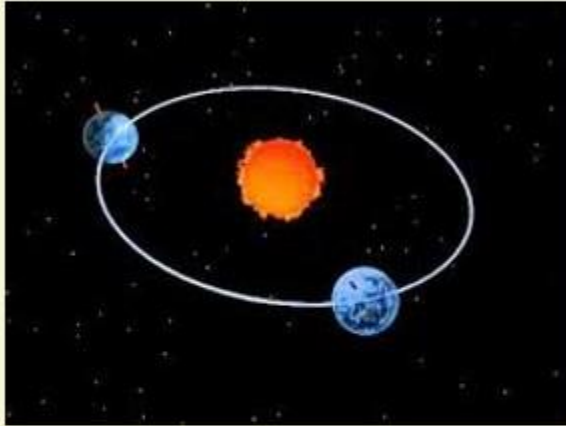
- 1** Rotation can be defined as the spinning of the Earth on its own axis.
- 2** Earth rotates from west to east around an imaginary line called axis.
- 3** Rotation causes day and nights.
- 4** Earth's rotation is completed in approximately 24 hrs.

REVOLUTION OF EARTH

- 1** Revolution can be defined as movement of Earth around the Sun.
- 2** Earth revolves around the Sun in an elliptical orbit, with the Sun as its focus.
- 3** Earth's axial tilt along with revolution causes different seasons.
- 4** Earth's revolution takes around 365 days.



Why the Earth is rotating?

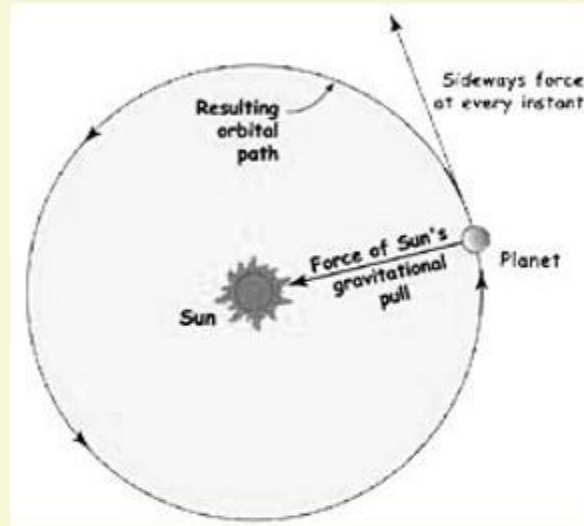


- The Earth was formed out of a nebula that collapsed. As the nebula collapsed, it began rotating.
- The Earth's rotation comes from the initial tendency to rotate that was imparted on it, when it was formed.
- However, for the things what we see on Earth, the force of friction tends to stop moving forever.
- But, for the Earth rotating on its axis, there is no force working to counteract the rotation.
- The only force available is tidal effect of the Moon, but that is very much insignificant.

The technical name we can infer is Angular Momentum. It is based on the principle that the total angular momentum of a system remains constant, unless acted upon by an external torque.



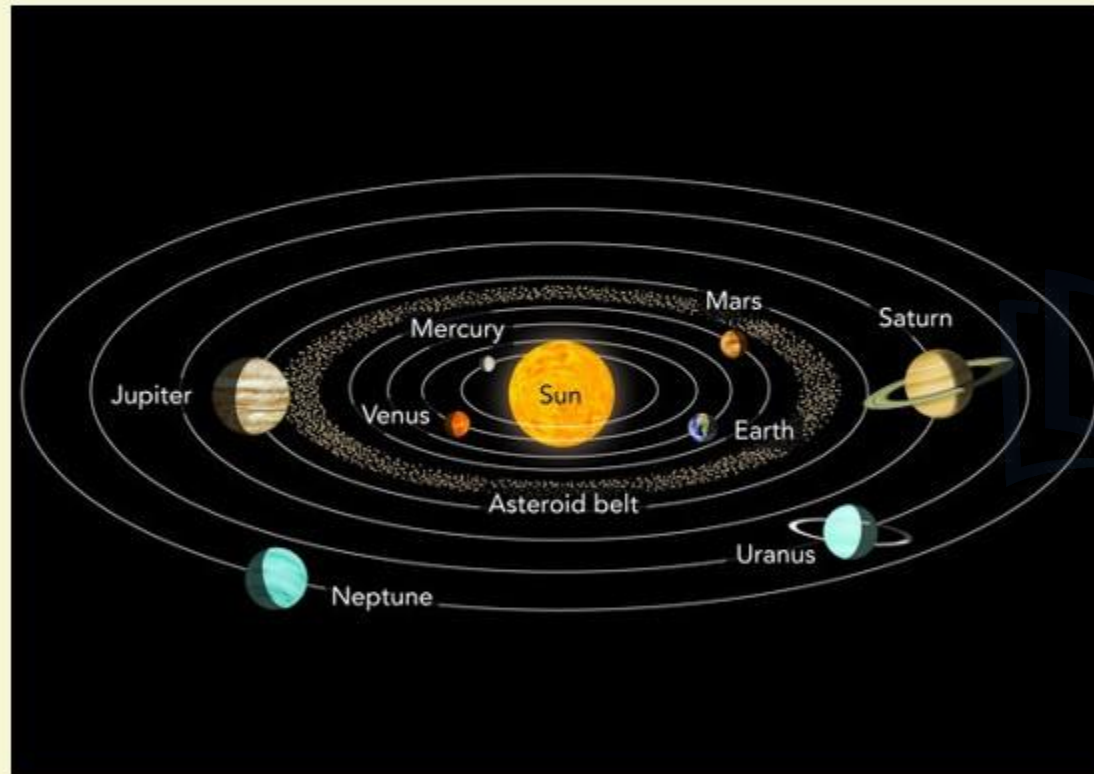
Why the planets revolve around the Sun?



- All of us very well know, Earth's gravity pulls down anything that is not held up by some other force.
- That is the reason, why all of us are standing/sitting on the ground.
- Heavier objects produce a bigger gravitational pull than the lighter ones.
- The Sun is the heavyweight in our solar system.
- It exerts the strongest gravitational pull.



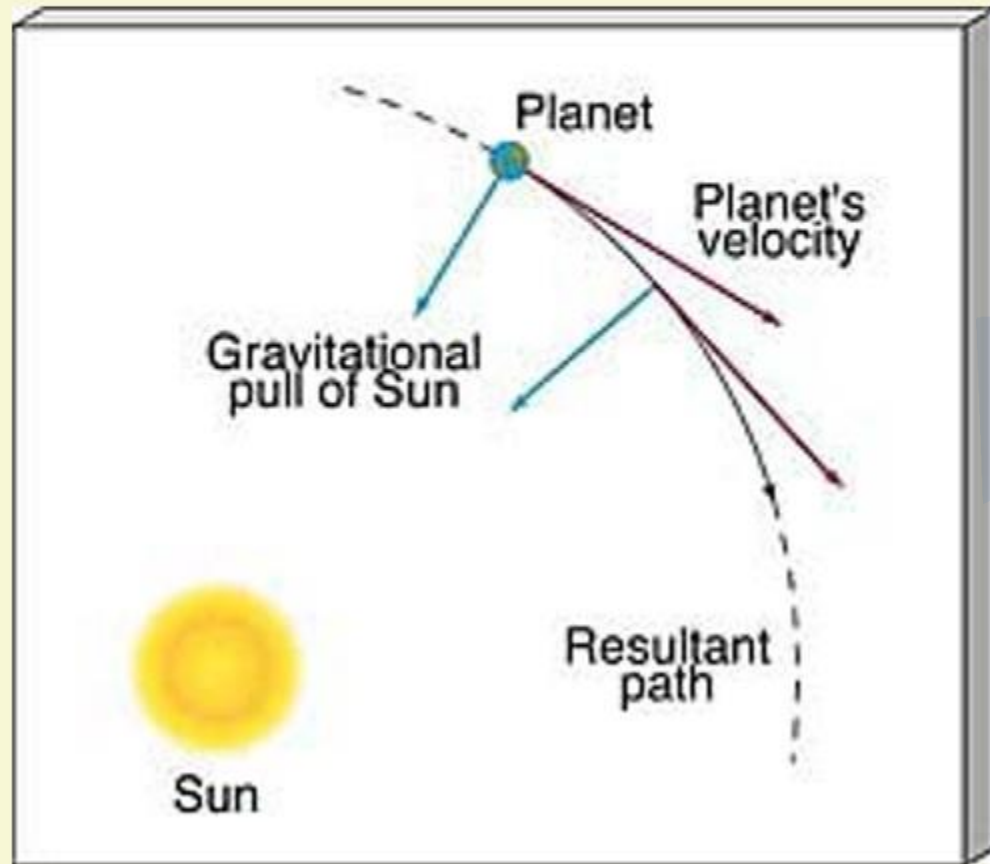
Why the planets revolve around the Sun? (... contd)



- Actually the planets are moving sideways and without that sideways motion, the planets would fall to the Centre, that is towards the Sun.
- Gravity of the Sun keeps them in their orbits.
- This is precisely how the **Moon orbits the Earth**, because of the pull of the **Earth's gravity**. Similarly, **Earth orbits the Sun**, because of the pull of the **Sun's gravity**.



Why the planets revolve around the Sun? (... contd)



MORE EXPLANATION ON THIS ...

- Earth has a velocity in the direction perpendicular to the force of the Sun's pull.
- If the Sun were not there, the Earth would travel in the straight line.
- But, the gravity of the Sun alters its course causing it to travel around the Sun, in a shape very near to a circle.

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